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BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401



Periodical Test - I - October 2025

S1 Semester

Answer Key**22MA101 & ENGINEERING MATHEMATICS I**

Degree & Branch: B.E./B.Tech. & CS, AL

Time: 1:30 hrs

Maximum: 50 Marks

Instructions:

1. Students should not mark/write anything on the Question Paper other than the register number.
2. Section A and B of the Question Paper contains questions for 30 Marks each. Section C contains questions for 15 Marks.
3. Students can attempt answering any two out of three subsections in each section. Maximum mark is limited to 20 in section A & B, and 10 in sections C.

Q. No.	Questions	
Section - A	COURSE OUTCOME 1	MAXIMUM: 20 MARKS
A1	(i)	d) 20
	(ii)	d) 55
	(iii)	The mathematical modelling is $y = a x + b$. From the method of least squares, the normal equations are $a \sum x + n b = \sum y$ $a \sum x^2 + b \sum x = \sum x y$ And, also $\sum x = 0$

$$\sum y = -1$$

$$\sum x = 2$$

$$\sum xy = 35$$

$$\sum y = 28$$

$$\sum x^2 = 30$$

$$\sum x^2 = 10$$

$$\sum xy = 36$$

Solving the normal equations, we have,

$$a = 3.5 \quad \text{and} \quad b = -0.2$$

$$a = 0.933$$

$$b = 6.5335$$

$$a = 0.7586$$

$$b = 6.6207$$

(iv)

The mathematical modelling is $y = ax + b$.

From the method of least squares, the normal equations are

$$a \sum x + nb = \sum y$$

$$a \sum x^2 + b \sum x = \sum xy$$

$$\text{And, also } \sum x = 10.$$

$$\sum y = 16.9$$

$$\sum xy = 47.1$$

$$\sum x^2 = 30$$

Solving the normal equations, we have,

$$a = 1.33 \quad \text{and} \quad b = 0.72$$

The mathematical model is $y = 1.33x + 0.72$

$$a = 0.0967 \quad b = 3.5734$$

A2

(i)

a) 6

(ii)

c) 38

(iii)

Given that the total current I is a linear combination of I_1 and I_2

$$I = aI_1 + bI_2$$

$$[2, 8] = a[4, 1] + b[-2, 2]$$

(1 Mark)

This gives the system of equations

$$4a - 2b = 2$$

$$a + 2b = 8$$

(1 Mark)

Solve the equations, we get

$$a = 2 \text{ and } b = 3$$

(1 Mark)

(iv)

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

(1 Mark)

Where $S_1 = 3$, $S_2 = 0$, $S_3 = -4$

(1 Mark)

Therefore the characteristic equation is $\lambda^3 - 3\lambda^2 + 4 = 0$

(1 Mark)

The Eigen values are -1, 2, 2.

The Eigen vectors are given by $(A - \lambda I)X = 0$

$$\text{For } \lambda = -1, \begin{pmatrix} 1 & -2 & -2 \\ -1 & 2 & 2 \\ -1 & -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

Therefore the corresponding Eigen vector is $\begin{pmatrix} 8 \\ 1 \\ 3 \end{pmatrix}$

(1 Mark)

$$\text{For } \lambda = 2, \begin{pmatrix} -2 & -2 & -2 \\ -1 & -1 & 2 \\ -1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

Therefore the corresponding Eigen vector is $\begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

For repeated root $\lambda = 2$, the Eigen vector $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ is same.

(1 Mark)

The largest Eigen value is 2 and the corresponding Eigenvector is $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$.

City A is likely a central hub or receives the highest traffic volume in the transportation network.

A3

(i)

a) 50

(ii)

c) 25

	(iii)	$\text{Mean } \bar{x} = \frac{\sum x}{n}$ $= \frac{9}{3}$ $= 3 \quad \text{----- (1 Mark)}$ $\text{Mean } \bar{y} = \frac{\sum y}{n}$ $= \frac{21}{3}$ $= 7 \quad \text{----- (1 Mark)}$ Regression Line x on y $x - \bar{x} = b_{xy}(y - \bar{y})$ $x - 3 = -0.5(y - 7)$ $x - 3 = -0.5y + 3.5$ $x = -0.5y + 3.5 + 3$ $x = -0.5y + 6.5 \quad \text{----- (1 Mark)}$	
	(iv)	Regression Line y on x $y - \bar{y} = b_{yx}(x - \bar{x})$ $y - 4 = 0.01(x - 250) \quad \text{----- (2 Mark)}$ $y - 4 = 0.01x - 2.5$ $y = 0.01x - 2.5 + 4$ $y = 0.01x - 1.5 \quad \text{----- (1 Mark)}$ The crop yield when 180 kg/ha of fertilizer is used is $y=3.3$. ----- (1 Mark) The crop yield when 340 kg/ha of fertilizer is used is $y=4.9$. ----- (1 Mark)	
Section - B		COURSE OUTCOME 2	MAXIMUM: 20 MARKS
B1	(i)	a)1955	
	(ii)	a) Upward	
	(iii)	The normal equations are $28a+7c=6$ $28b=40$ (1 Mark)	

$$196a + 28c = 6$$

On solving the above linear equations, we get $a = 1.4286$, $b = -0.21429$ and $c = 1.7143$.
(1 Mark)

The required polynomial to represent the growth of a plant is $y = 1.4286x^2 - 0.21429x + 1.7143$.
(1 Mark)

(iv)

a) The x-intercepts are the real solutions of the equation $p(x) = 0$.

$$\Rightarrow -2x^2 + 80x - 600 = 0$$

$$\Rightarrow x = 10 \text{ or } x = 30$$

(1 Mark)

The x-intercepts are (10, 0) and (30, 0).

$$p(0) = -600$$

The y-intercept is (0, -600).

(1 Mark)

b) The x-coordinate of the vertex is $h = \frac{-b}{2a} = 20$

The y-coordinate is

$$k = p(20) = -2(20)^2 + 80(20) - 600 = 200$$

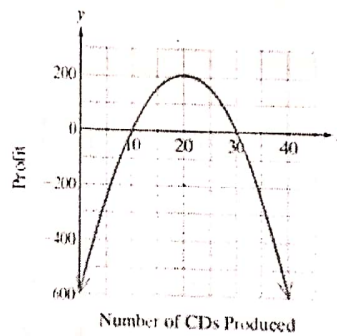
The vertex is (20, 200).

(1 Mark)

A maximum weekly profit of \$200 is obtained when 20 CDs are produced.

(1 Mark)

c).



(1 Mark)

B2

(i)

b)

$$\begin{bmatrix} 2 & -3 \\ -3 & 3 \end{bmatrix}$$

(ii)

c) 2

	(iii)	a) The given matrix is upper triangular matrix, the eigen values are 3, -1, 2 (1 mark) b) The canonical form the stability matrix is $3y_1^2 - y_3^2 + 2y_4^2$ (1 mark) c) Rank = 3 (1 mark) Nature = Indefinite (1 mark)
	(iv)	a) The matrix of the quadratic form is $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ (1 mark) b) The characteristic equation is $\lambda^3 - 0\lambda^2 - 3\lambda + 2 = 0$ (1 mark) The Eigen values are $\lambda_1 = -1, \lambda_2 = -1, \lambda_3 = 2$. (1 mark) Nature = Indefinite. (1 mark) Complex situation with trade-offs and the potential for saddle points, where the firm must carefully balance inputs.
B3	(i)	d) Partial definite
	(ii)	a) A^T
	(iii)	a) The matrix of the quadratic form is $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ (1 mark) b) The characteristic equation is $\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$ (1 mark) The eigen values are $\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 8$. (1 mark) Nature = positive definite. the robot exhibits stable motion (1 mark)
	(iv)	When $\lambda = 2$, the corresponding eigen vector is $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. -----(1m) The normalized modal matrix $N = \begin{pmatrix} \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}$ -----(1m) $N^T = \begin{pmatrix} \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}$ -----(1m) $N^T A N = D = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$ -----(1m)
Section - C		COURSE OUTCOME 3
C1	(i)	c) 4
	(ii)	

$\log_{10}(I)$	$\log_{10}(H)$
0.3010	0.6990
0.6021	0.9542
0.7782	1.1761
0.9031	1.3802
1.0000	1.5441

----- (1)

$$\sum Y = 5.7536; \sum X = 3.5844; \sum X^2 = 2.8743; \sum XY = 4.4907 \text{ ----- (2)}$$

The normal equation is |

$$5A + 3.5844 b = 5.7536; \quad 3.5844 A + 2.8743 b = 4.4907 \text{ ----- (1)}$$

C2

(i)

c) a=2.5, b=10

(ii)

$$12.7 = 5a + 10b + 30c + 100d$$

$$36.5 = 10a + 30b + 100c + 354d$$

$$128.5 = 30a + 100b + 354c + 1300d$$

$$477.5 = 100a + 354b + 1300c + 4890d$$

----- 4 marks

C3

(i)

c) y increases more slowly than x

(ii)

On solving the normal equations, we get

$$A = 1.967 \text{ ----- (1)}$$

$$b = 1.311 \text{ ----- (1)}$$

$$A = \log_{10}(a) \Rightarrow a = 10^A \Rightarrow a = 10^{(1.967)} \Rightarrow a = 92.683 \text{ ----- (1)}$$

$$\text{The power curve is } P = (92.683)^{1.311} \text{ ----- (1)}$$

*** End of Question Paper ***